16.05.17(CE)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
End Semester Exam

Max. Marks: 100	**April-May 2017**	**Duration: 3 Hrs**
Class: S Y BTech		Semester: IV
Name of the Course: Applied Mathematics-IV		Branch: COMP/IT
Course Code: UCEC401 / UITC401		

Instructions:
(1) All Questions are Compulsory
(2) Figures to right indicate full marks.

Question NO.		Questions	Marks
Q.1	A	The Probability density function of a random Variables x is represented by the following $f(x) = 2x^3 \quad 0 \le x \le 1$, $= 2(2-x)^3 \quad 1 \le x \le 2$. Find mean and standred deviation of random variables x.	5
	B	Attempt any Two of the following.	16
	i	Heights of father and son are given below in centimeters. Height of father: 150, 152, 155, 157, 160, 161, 164, 166 Height of son: 154, 156, 158, 159, 160, 162, 161, 164 Find the line of regression of height of father on the height of son. Also calculate the expected height of the father when the height of son is 157cm.	
	ii	If x and y are two correlated variables with same standared deviation and the correlation coefficient ρ then find the correlation coefficient between x and x +y	
	iii	Fit a Parabola to the following data and estimate $y(6)$ x: 1, 2, 3, 4, 5 y: 25, 28, 33, 39, 46	
Q.2	A	The amount of time that awatch will run without having to be reset is a random variable having an exponential distribuation with mean 120 days. Find the probability that such a watch will have to be set in less than 24 days.	5
	B	Attempt any Two of the following.	16
	(i)	If on an average rain falls on 10 days in every 30 days obtain the probability that I) rain will fall on at least 4 days of a given week. II) first 4 days of a given week will be fine and the remaining 3 days wet.	
	(ii)	A distributer of an seeds determineds from extensive tests that 5 % of large batch of seeds will not germinate.He sells seeds in packeds of 200 snd guarantees 90 % germination. Determine the probability that a particular packet will violate the guarantee.	
	(iii)	The Marks x obtained in mathematdics by 1000 students is normally distributed with mean 78 % and Standerd deviation 11 %. I) Determine how many students got marks above 90 %. II) What was the highest mark obtained by the lowest 10 % of students.	

Q.3

A A sample of 100 students is taken from a large population. The mean height of these student is 64 cm and Standred deviation is 4 cm. Can it be reasonably regarded as drawn from the population with mean height 60 cm ? — **5**

B Attempt any Two of the following. — **16**

i) A drug was administered to 5 persons and the systolic blood pressure before and after was measured. The result are given below

Candidates	I	II	III	IV	V
B.P. before	140	130	132	150	140
B.P. after	132	126	133	144	133

Test whether the drug is effective in lowering the systolic blood pressure. (Take 1% level of significance)

ii) According to theory the proportion of a commodity in the 4 classes A, B, C, D should be 9 :4 : 2: 1. In a survey of 1600 items of this commodity the numbers in the four classes were 882, 432, 168, & 118. Does the survey support the theory?

iii) Two kinds of manures were used in sixteen plots of the same size other conditions being the same. The yields in quintals are given below.

Manure I	18	20	36	50	49	36	34	49	41
Manure II	29	28	26	35	30	44	46		

Test at 5% level of significance whether the two manures differ as regards their mean yields.

Q.4 Attempt any TWO of the following. — **16**

(i) A diesel pump has capacity to accommodated only 4 trucks including the one at pump. On an average trucks arrive at the rate of 5 per hr and each truck take 10 min for service. Assume that the arrival process is Poisson and the service time is an exponential random variable.

I) What is the average time for which a truck is at pump?

II) What is the average waiting time for for a truck?

III) Find the percentage of trucks that will turn away?

(ii) Passengers arrive at local station at the rate of 6 per min and a passenger needs 7.5 sec to get ticket.Mr.X arrives 2 min before the local starts and it takes exactly 1.5 min for him to get into local after purchasing the ticket.

I) Can he be in the local before the local starts?

II) what is the probability that he will be in the local before it starts

III) If the probability that he will get into local is to be 0.99, how early must be arrive?

(iii) Customers arrive at a clinic according to a Poisson process with a mean interval of 25 min. The Physician needs on an average 20 min for a patient to examine.

I) find the expected number of patients at the clinic.

II) Find the percentage of patients whi ch are not required to wait.

III) On an average how much time is spent by a patient in the clinic?

IV)The Physician will appoint another physician (equally competent) If the patient's time in the clinic exceeds 2 hr. How much must the rate of arrivals increase so that another physician is appointed?

Q.5	A	Find the Maximum or minimum of the function $Z = x_1 x_2 + 9x_2 + 6x_3 - x_1^2 - x_2^2 - x_3^2$	5
	B	Attempt any Two of the following.	16
	i	Solve the following L.P.P. by Simplex method Maximise $z = 5x_1 + 4x_2$ Subject to $6x_1 + 4x_2 \le 24$ $x_1 + 2x_2 \le 6$ $-x_1 + x_2 \le 1$ $x_2 \le 2$ where $x_1 \ge 0, x_2 \ge 0$	
	ii	Using the method of Lagrange's Multiplier solve the following N.L.P.P. Optimize $z = 10x_1 + 8x_2 + 6x_3 + 2x_1^2 + x_2^2 + 3x_3^2 - 100$ Subject to $x_1 + x_2 + x_3 = 20$ where $x_1, x_2, x_3 \ge 0$	
	iii	Using Big M method solve the following LPP Minimize $z = 2x_1 + x_2$ Subject to $3x_1 + x_2 = 3,$ $4x_1 + 3x_2 \ge 6$ $x_1 + 2x_2 \le 3$ $x_1, x_2 \ge 0$	